[Def. Define

$$f:[0,1]\to\mathbb{R}:x\mapsto\begin{cases}\frac{1}{p} & \text{if } x\in\mathbb{Q} \text{ with } x=\frac{q}{p}, \text{ irreducible form,}\\ 0 & \text{otherwise.}\end{cases}$$

Then,
1). f is continuous at every irrational numbers, and discontinuous at every rational numbers.
2) f is Riemann-integrable.
Prof. observe that for each $n\in\mathbb{N}$, the set

$$S_n=\left\{x\in[0,1]\,\middle|\, f(x)>\frac{1}{n}\right\}=f^{-1}\left[\left(\frac{1}{n},\infty\right)\right]=\left\{\frac{q}{p}\in[0,1]\,\middle|\,\gcd(p,q)=1,\ 1\le p<n\right\}.$$

is finite set,
Let $x_0\in[0,1]$ be an irrational number. $\min\{|x_0-y|\,|\,y\in S_N\}$.
Given $\varepsilon>0$, there is $N\in\mathbb{N}$ s.t $\frac{1}{N}<\varepsilon$. Since the S_N is finite, we can take $\delta>0$ s.t

$$x\in(x_0-\delta,x_0+\delta)\Rightarrow|f(x_0)-f(x)|=f(x)\le\frac{1}{N}<\varepsilon.$$

So, f is continuous at any irrational. And, for rational $\frac{q}{p}\in[0,1]$,

for any $\delta>0$, there is an irrational $x\in\left(\frac{q}{p}-\delta,\frac{q}{p}+\delta\right)$ s.t $\left|\frac{1}{p}-f(x)\right|=\frac{1}{p}>\frac{1}{2p}$.

So it is not continuous at rational.] 1).

Given $\varepsilon>0$, let $N\in\mathbb{N}$ be a number such that $\frac{1}{N}<\frac{\varepsilon}{2}$.
Denote that $S_N=\{x_1,\cdots,x_k\}$. Since this S_N is finite, put $d>0$ such that

$$d<\min\{|x_i-x_j|\,|\,i\neq j\} \quad\text{and}\quad d\cdot k<\frac{\varepsilon}{2}.$$

Now, construct a partition on $[0,1]$ as

$$P=\left\{0,1,x_1-\frac{d}{2},x_1+\frac{d}{2},\cdots,x_k-\frac{d}{2},x_k+\frac{d}{2}\right\}\cap[0,1].$$

Then, by the density of irrationals, the Lower sum $L(f,P)=0$, and

$$U(f,P)\le 1\cdot\frac{1}{N}+1\cdot d\cdot k$$

Now, $\overline{\int}f-\underline{\int}f\le U(f,P)-L(f,P)\le\frac{1}{N}+dk<\frac{\varepsilon}{2}+\frac{\varepsilon}{2}$

gives the result, by the integrable criterion. □